

Gosala Ramaiah

Lead Engineer- Hardware Design & Development

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Objective	As Lead Hardware Engineer with overall 8.8 years of experience in hardware design, schematic development, 4-layer PCB design on digital systems testing, seeking a challenging role to lead end-to-end product development and deliver robust, high-quality hardware solutions.
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Education	[JNTU-Kakinada University-Guntur Engineering Collage] [2011] – [2015] B. Tech in Electronics and communication engineering
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Key Skills	Schematic Design-Altium 23.3.1 1,2,4 layers PCB design. Testing power on & functional.	Troubleshoot. Quality. Communication.
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Experience	Responsibilities: Managed schematic design activities, PCB design- footprint creation, component placement, PCB routing. Troubleshoot routing-related issues. Performed ESS, Oscilloscopes, Multimeter, function generators, spectrum analyzers. Quality report preparation, Engage in discussions. Perform functional test. PIC, ESP32, STM32, FPGA.	
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March 2025 - Present	July 2019 - April 2024	July 2016 – June 2019
Lead Engineer-Hardware D&D.	Quality and Hardware Testing. Sr Hardware Testing. Design Engineer.	Trainee, Hardware Testing Engineer.
[GANDEEVAN TECHNOLOGIES (OPC)PVT LTD]	[ELDAAS/THRISHANKU TECHNOLOGIES PVT LTD]	[NSK ELECTRONICS]

Enhancement	Java Core Training [oct 2015-April 2016] – Aradhya’s Brilliance Centre, Bangalore <i>Completed 6-month training program on core Java.</i> Career Break: [May 2024 – March 2025] • Took a planned break due to personal health reasons. Now fully recovered and ready to resume professional work.
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Tools	Altium 23.3.1, (schematic and layout), ORCAD 17.4(schematic) Power Supply, Oscilloscopes, Digital Multimeter, function generators, spectrum analyzers.
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Projects Overview:

Project-1: Role: Schematic Design and layout for Smart PITS modules.

Scope of Work: The Smart PITS module is designed to detect and respond to even small vibrations by using Accelerometer, Gyroscope. Whenever a vibration is detected, the system captures the vibration data and transmits it to another module. The schematic and PCB layout are developed using the STM32L431 microcontroller. For communication between modules, the CAN protocol is implemented.

Project-2: Role: Schematic Design and layout for STM32/ESP32 modules.

Scope of Work: The STM32/ESP32 development boards are designed with a complete schematic and a 4-layer PCB layout to ensure high performance and reliability. Making the board suitable for complex and high-performance embedded system development.

Project-3: Role: Testing Engineer for FCC &PDU modules.

Scope of Work: Involves in functionality testing modules used in drone and flight systems. The FCC module is used for controlling signals in flying devices and drones. Testing includes RS-232 and RS-422 communication, Ethernet, CAN, PWM, ADC, LOAD using Xilinx SOM module. The PDU is responsible for distributing equal power. Performed ESS (Environmental Stress Screening).

Project -4: Role: Testing Engineer dual-channel detector DCD

Scope of Work: Involves in functionality testing DCD module, which has two channels (A and B), was tested using a function generator to provide input signals at fixed frequencies of 20 MHz and 30 MHz the input power levels varied from +10 dBm to -70 dBm. For each frequency and power level, the binary output data from the DCD module was captured and verified against the expected write data to ensure accuracy.

Project -5: Role: Design Engineer of Back Panel

Description: updated schematic back-panel card for computer and networking data transfer features two 110-pin CPCI connectors, designated as Front and Back Panels (J1 and J2), for handling input and output signals. Input signals are routed directly to a Kintex-7 FPGA through the Front Panel, allowing high-speed processing and control. On the output side, the FPGA communicates with a 16-bit ADC in Cluster-1 via an SPI interface, with each data line protected using digital isolators to prevent noise and ground-loop interference.

Project -6: Role: Testing of MAT Power Supply Card Prototype

Scope of Work: Involves in functionality testing modules The MAT power supply is designed for autonomous underwater vehicles, operating with an input of 48 V at 5 A. The testing process includes verifying all power outputs, evaluating the RS-422 output, and conducting load tests at different intervals, such as 5 and 10 minutes, to assess performance under normal operating conditions. Additionally, the prototype undergoes environmental stress testing (ESS) to ensure reliable operation under varying underwater conditions.

Project -7: Role: Testing K9 Vajra Tank Units

Description: The work involves functional testing of critical modules for the K9 Vajra Tank, including the power module, joystick controller, analog and filter module, DIP, ACU, relay module, and motherboard. Each of these components is essential to ensure the tank's overall functionality and operational performance in the field. The process includes thorough quality verification, preparation of Test Data Reports (TDRs), and documentation of quality standards. In addition, external quality and functional testing of the cards is conducted in collaboration with a third-party vendor to validate compliance with specifications and ensure reliability under operational conditions. This comprehensive approach ensures that all modules perform correctly and meet stringent quality requirements.

Project -8: Role: Testing Battery Operating Current Source (BOCS).

Description: As a Product Testing Engineer for the Battery Operating Current Source (BOCS) project, my primary role is to ensure the reliability and functionality of this essential component, which finds applications in industrial settings like atomic emission spectrometers and material analyzers. The BOCS plays a critical role in identifying and measuring the purity of materials, including precious metals like gold. It operates within specified over voltage and under voltage conditions and communicates with the system using CAN communication protocols. operate the input gas as a spark source, and we employ Zig test controller to monitor and limit gas usage, optimizing the performance of the system.

I hereby declare that I possess expertise in Electronics and Communication Engineering and have hands-on experience in embedded systems, hardware design, and project development. I confirm that the details provided in this resume are accurate to the best of my knowledge.

Place: Bengaluru,
Date:08.04.2026.

Name : Gosala .Ramaiah